

Student: Toheeb Eji
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Instructor: Aleksandra Gajic
Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 8

1. Into which category of aggregate expenditures¹ would the following transaction fall?

The Jones family buys a new car. Consumption expenditure

1: More Info

Aggregate expenditure (AE) is the total amount of spending in the economy: the sum of consumption, planned investment, government purchases, and net exports.

2. Suppose Apple plans to produce 2.4 million iPhones in Canada this year. The company expects to sell 2.25 million and add 150,000 to the inventories² in its stores.

a. Suppose that at the end of the year, Apple has sold 1.85 million iPhones. What was Apple's planned investment? What was Apple's actual investment?

b. Now suppose that at the end of the year, Apple has sold 2.65 million iPhones. What was Apple's planned investment? What was Apple's actual investment?

a. Apple's planned investment was million iPhones. (Round your response to two decimal places.)

Apple's actual investment was million iPhones. (Round your response to two decimal places.)

b. Apple's planned investment was million iPhones. (Round your response to two decimal places.)

Apple's actual investment was million iPhones. (Round your response to two decimal places.)

2: More Info

Inventories are goods that have been produced but not yet sold.

3. Which of the following is not an explanation for why firms keep inventories?

- ☐ A. Firms keep inventories in order to provide customers with a wider selection of choices.
- ☒ B. Because goods are produced in quantities that match the exact amounts consumers want to purchase at a given time.
- ☐ C. Inventories allow firms a place to store excess production if they incorrectly estimate the quantity that consumers wish to purchase.
- ☐ D. Because customers do not purchase goods at regular intervals, inventories allow firms to have goods in stock when customers want to buy.
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4. On the graph of the consumption function, the horizontal axis measures _____, while the vertical axis measures _____.

- ☐ A. real GDP; real aggregate expenditure
- ☐ B. planned consumption spending; real national income
- ☐ C. real disposable income; real national income
- ☒ D. real national income or real GDP; real consumption spending

The graph on the right shows a consumption function, C_1 . The current income-consumption combination is shown by point A.

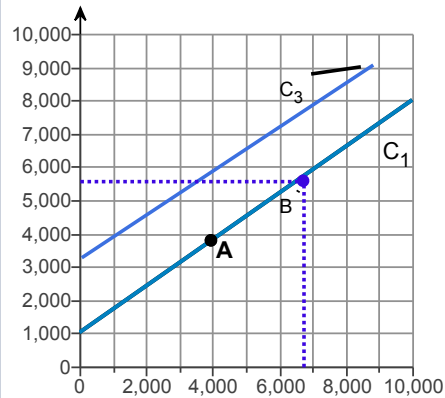
Suppose the level of income increases.

1.) Use the point drawing tool or the line drawing tool (as appropriate) to show the impact of the increase in income on consumption spending. Label the new income-consumption combination 'B' or the new consumption function, if any, ' C_2 '.

Suppose there is an increase in household wealth.

2.) Use the point drawing tool or the line drawing tool (as appropriate) to show the impact of such an increase. Label the new income-consumption combination 'D' or the new consumption function, if any, ' C_3 '.

Carefully follow the instructions above, and only draw the required objects.



5. Which of the following is NOT a determinant of net exports?

- ☐ A. The growth rate of the U.S. price level relative to price levels of other countries
- ☐ B. The exchange rate between the dollar and other currencies
- ☒ C. The interest rate
- ☐ D. The growth rate of U.S. GDP relative to growth rates of other countries.

6. Complete the following table to indicate what effect an increase in each of the consumption components will have on consumption.

An increase in...	will (increase/decrease)	consumption.
the price level	decrease	consumption.
household wealth	increase	consumption.
expected future income	increase	consumption.
current disposable income	increase	consumption.
the interest rate	decrease	consumption.

7. If the marginal propensity to consume is 0.60, a \$50 increase in disposable income will lead to a \$ increase in consumption and a \$ increase in savings. (Enter both responses rounded to two decimal places.)

8. Which of the following will increase planned investment spending on the part of firms?

- ☐ A. A lower real interest rate
- ☐ B. Increased optimism about future demand for its product
- ☐ C. Increases in the corporate income tax
- ☐ D. All of the above
- ☒ E. A and B only

9. What is the meaning of the 45° line in the 45°-line diagram?

In the 45°-line diagram, the 45° line shows

- ☐ A. real consumption spending for varying levels of real GDP.
- ☐ B. real aggregate expenditure for varying levels of real GDP.
- ☒ C. all the points where aggregate expenditure equals real GDP.
- ☐ D. the equality of real consumption spending and real GDP.

10. What does the slope of the aggregate expenditure³ line equal? How is the slope of the aggregate expenditure line related to the slope of the consumption function⁴?

The slope of the aggregate expenditure line is $\frac{\Delta AE}{\Delta \text{Real GDP}}$. The simple aggregate expenditure model assumes that taxes, planned investment spending, government purchases, and net exports are independent of income. So, the slope of the aggregate expenditure line equals the slope of the consumption function, which is the marginal propensity to consume.

3: Definition

Aggregate expenditure (AE) is total spending in the economy: the sum of consumption, planned investment, government purchases, and net exports.

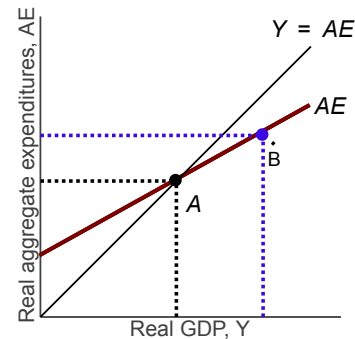
4: Definition

The consumption function is the relationship between consumption spending and disposable income.

11. The diagram shows a 45°-line (Keynesian cross) diagram.

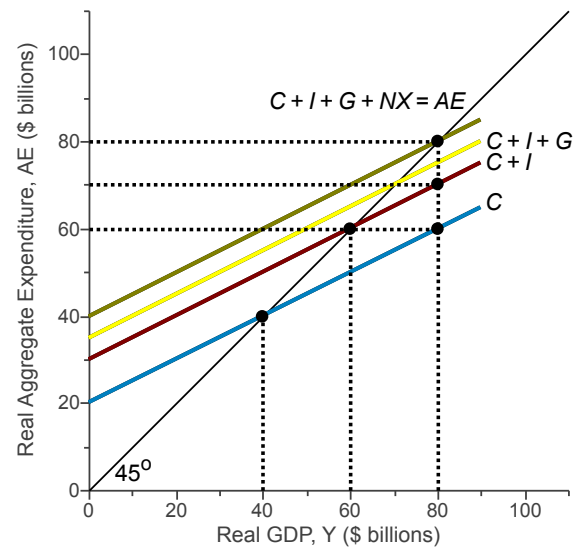
Use the point tool to select a point *on the expenditures line* where: inventories are rising. Label this point 'B'.

Carefully follow the instructions above, and only draw the required object.



12. Consider the figure. This economy is in macroeconomic equilibrium at what level of real GDP? \$ billion.

What is the level of planned investment? \$ billion.



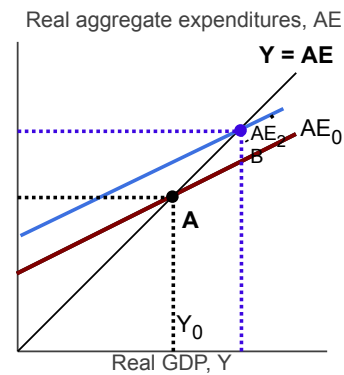
13. The graph below shows a 45°-line (Keynesian cross) diagram. The economy is currently in macroeconomic equilibrium at output level Y_0 .

Suppose that investment increases.

- 1) Use the line tool to show a possible position for the new aggregate expenditures line. Label this line AE_2 .

Note: if you are not prompted for a label, you have used the wrong drawing tool.

- 2) Use the point drawing tool to show the new equilibrium levels of GDP and expenditures. Label this point 'B'.



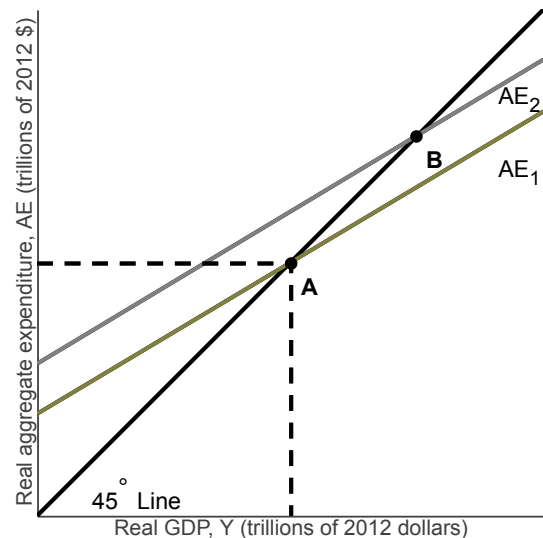
14. In the graph to the right, the economy is initially in equilibrium at point A. Aggregate expenditure and real GDP both equal \$9.6 trillion. Suppose there is an increase in investment spending of \$100 billion that increases aggregate expenditure to \$9.7 trillion.

Which of the following best describes the initial impact of the increase in investment?

- ☐ A. Initially, real GDP rises by \$100 billion.
- ☐ B. The aggregate expenditure line shifts up by \$100 billion.
- ☐ C. The economy is no longer in equilibrium.
- ☒ D. All of the above.

All of the following are true as the economy adjusts to a new equilibrium *except that*

- ☒ A. the initial increase in income or GDP leads to a further increase in investment and aggregate expenditure.
- ☐ B. the initial increase in income or GDP leads to a further increase in consumption and aggregate expenditure.
- ☐ C. the final increase in GDP will be larger than \$100 billion.
- ☐ D. the economy will arrive at an equilibrium at point B.



15. If the marginal propensity to consume is 0.50, by how much will an increase in planned investment spending of \$60 billion shift up the aggregate expenditure line?

If planned investment spending increases by \$60 billion, it will shift the aggregate expenditure line up by \$ billion. (Round your response to the nearest whole number.)

By how much will it increase equilibrium real GDP?

Equilibrium real GDP will increase by \$ billion. (Round your response to the nearest whole number.)

16. Would a larger multiplier⁵ lead to more severe recessions or less severe recessions?

- ☐ A. A larger multiplier means that small changes in spending lead to large changes in GDP, and thus recessions would be less severe
- ☐ B. A larger multiplier means that large changes in spending lead to small changes in GDP, and thus recessions would be less severe
- ☒ C. A larger multiplier means that small changes in spending lead to large changes in GDP, and thus recessions would be more severe
- ☐ D. A larger multiplier means that large changes in spending lead to small changes in GDP, and thus recessions would be more severe

5: Definition

Multiplier The increase in equilibrium real GDP divided by the increase in autonomous expenditure.

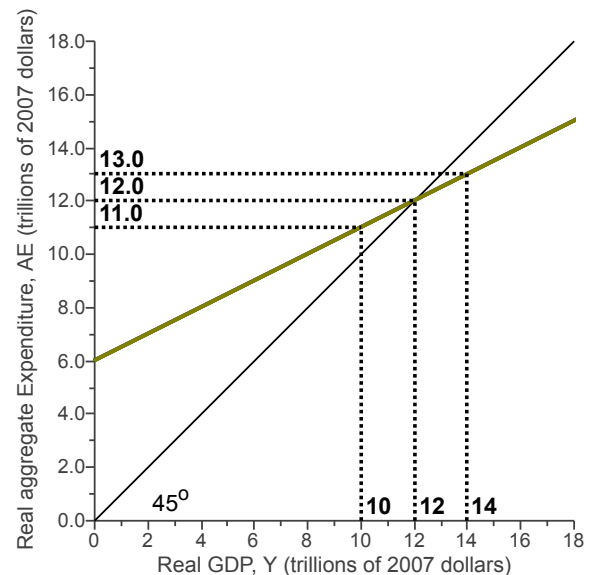
17. Use the diagram to the right to answer the following.

- a. What is the value of equilibrium real GDP?
- b. What is the value of the MPC?
- c. What is the value of the multiplier?

a. The equilibrium value of real GDP is \$ trillion. (Round your response to the nearest whole number.)

b. The MPC⁶ is equal to . (Round your response to two decimal places.)

c. The multiplier⁷ is equal to . (Round your response to one decimal place.)



6: More Info

Marginal propensity to consume (MPC) represents the slope of the consumption function. It shows the amount by which consumption spending changes when disposable income changes.

7: More Info

Multiplier is the increase in equilibrium real GDP divided by the increase in autonomous expenditure.

18. In a closed economy, consumers spend \$300 regardless of the level of income, and the marginal propensity to consume (MPC) is 0.75. Investment is equal to \$350. The government spends \$300 and collects \$150 in taxes.

The equilibrium level of GDP in this economy is \$ (enter your response rounded to the nearest dollar).

19. Suppose the marginal propensity to consume increases from 0.67 to 0.85.

The value of the expenditure multiplier will increase from to . (Enter your responses rounded to two decimal places.)

When the value of the multiplier increases, all else equal, a change in expenditure will raise aggregate expenditure by a larger amount.

20. Briefly explain whether the following statement is true. "The reason that the aggregate demand⁸ curve slopes downward is that when the price level is higher, people cannot afford to buy as many goods and services."

The statement is true. An increase in the price level causes consumption, planned investment, and net exports to fall, which decreases real GDP.

8: Definition

The aggregate demand (AD) curve is a curve that shows the relationship between the price level and the level of planned aggregate expenditure in the economy, holding constant all other factors that affect aggregate expenditure.

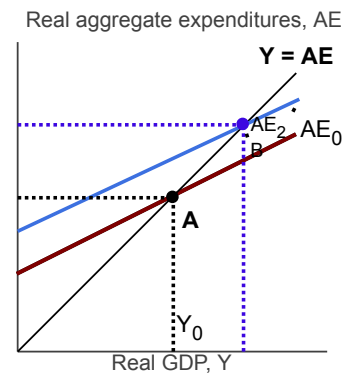
21. The graph to the right shows a 45°-line (Keynesian cross) diagram. The economy is currently in macroeconomic equilibrium at output level Y_0 .

Suppose that the price level decreases.

- 1) Use the line tool to show a possible position for the new aggregate expenditures line. Label this line AE_2 .

Note: if you are not prompted for a label, you have used the wrong drawing tool.

- 2) Use the point drawing tool to show the new equilibrium levels of GDP and expenditures. Label this point 'B'.



22. Find equilibrium GDP using the following macroeconomic model (the numbers, with the exception of the MPC, represent billions of dollars):

$C = 1,500 + 0.80Y$	Consumption function
$I = 1,000$	Planned investment function
$G = 2,000$	Government spending function
$NX = -500$	Net export function
$Y = C + I + G + NX$	Equilibrium condition

The equilibrium level of GDP is \$ billion. (Round your answer to the nearest billion dollars.)

23. Consider the macroeconomic model shown below:

$C = 150 + 0.80Y$	Consumption function
$I = 150$	Planned investment function
$G = 100$	Government spending function
$NX = -50$	Net export function
$Y = C + I + G + NX$	Equilibrium condition

Fill in the following table. (Round your responses to the nearest dollar.)

GDP	Aggregate Expenditures (AE)	Unplanned Change in Inventories
\$1,400	\$ 1470	\$ -70
\$2,100	\$ 2030	\$ 70

24. Suppose that autonomous consumption is 50, government purchases are 150, planned investment spending is 200, net exports are - 20, and the MPC is 0.6.

Equilibrium GDP is \$. (Round your response to the nearest dollar.)

25. Fill in the blanks in the following table. Assume for simplicity that taxes are zero. Also assume that the values represent billions of 2012 dollars. (Round your responses to the nearest whole number.)

Real GDP (Y)	Consumption (C)	Saving (S)
\$900	\$720	\$ 180
\$1,000	\$800	\$ 200
\$1,100	\$880	\$ 220
\$1,200	\$960	\$ 240
\$1,300	\$1,040	\$ 260

The marginal propensity to consume⁹ is . (Round your response to two decimal places.)

The marginal propensity to save¹⁰ is . (Round your response to two decimal places.)

9: Definition

Marginal propensity to consume (MPC) represents the slope of the consumption function. It shows the amount by which consumption spending changes when disposable income changes.

10: Definition

Marginal propensity to save (MPS) is the change in saving divided by the change in disposable income.

26. a. Fill in the missing values in the following table. Assume that the value of the MPC ¹¹ does not change as real GDP changes and that there are zero taxes. Also assume that the values represent billions of 2012 dollars. (Round your responses to the nearest whole number.)

Real GDP (Y)	Consumption (C)	Planned Investment (I_p)	Government Purchases (G)	Net Exports (NX)	Planned Aggregate Expenditure (AE)	Unplanned Change in Inventories
\$1,500	\$1,350	\$100	\$100	- \$40	\$ 1510	\$ - 10
1,600	1,440	100	100	- 40	1600	0
1,700	1530	100	100	- 40	1690	10
1,800	1620	100	100	- 40	1780	20
1,900	1710	100	100	- 40	1870	30

The value of the MPC is . (Round your response to two decimal places.)

- b. The value of equilibrium real GDP is \$ billion. (Round your response to the nearest whole number.)

11: More Info

Marginal propensity to consume (MPC) represents the slope of the consumption function. It shows the amount by which consumption spending changes when disposable income changes.

27. Use the information in the table below to answer the following questions. Assume that the values represent billions of 2012 dollars.

Real GDP (Y)	Consumption (C)	Planned Investment (I)	Government Purchases (G)	Net Exports (NX)
\$600	\$350	\$200	\$250	– \$50
\$700	\$400	\$200	\$250	– \$50
\$800	\$450	\$200	\$250	– \$50
\$900	\$500	\$200	\$250	– \$50
\$1,000	\$550	\$200	\$250	– \$50

a. The equilibrium level of real GDP is \$ billion. (Round your response to the nearest whole number.)

b. The MPC^{12} is . (Round your response to two decimal places.)

c. Suppose that net exports increase by \$20 billion. Using the multiplier formula, determine the new level of real GDP.

A \$20 billion increase in net exports leads to a change in spending of \$ billion, so the new level of real GDP will be \$ billion. (Round your responses to the nearest whole number.)

12: Definition

Marginal propensity to consume (MPC) represents the slope of the consumption function. It shows the amount by which consumption spending changes when disposable income changes.

28. Suppose booming economies in developing countries cause net exports (NX) to rise by \$75 billion in Canada.

If the MPC^{13} is 0.5, the change in equilibrium GDP will be \$ billion. (Round your response to the nearest whole number.)

13: Definition

Marginal propensity to consume (MPC) is the amount by which consumption spending changes when disposable income changes. It is also equal to the slope of the consumption function.